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Firing Devices

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

Describes procedures for testing demolition-initiating equipment. Includes explosive and nonexplosive items used to detonate explosive charges.

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# APPENDIX B

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

AMSTE-RP-702-103
*Test Operations Procedure 4-2-045 26 February 1985
AD No. A151272

# DEMOLITION-INITIATING EQUIPMENT


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1. SCOPE. This TOP provides guidance for planning tests of demolition-initiating equipment to assure conformance with requirements documents. Tests to satisfy the requirements for the particular test item and test type can be selected from those listed in paragraph 4. This procedure does not apply to firing devices (fuzes) as defined in Appendix A, to initiating components of special demolition items or kits, or to their tools and accessories. This TOP does not cover mines and demolitions; for testing procedures regarding these items, see TOP 4-2-505.[1] **

Demolition-initiating equipment consists of the explosive and nonexplosive items used to initiate standard or expedient demolition explosive charges.

*This TOP supersedes TOP 4-2-045 dated 2 February 1972.

**Footnote numbers correspond to reference numbers in Appendix B.

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# APPENDIX A
# CATEGORIES OF DEMOLITION-INITIATING EQUIPMENT

## 1. Blasting Machine

The blasting machine is a manually powered generator for initiating electric blasting caps. Blasting machines are rated according to the number of electric blasting caps, connected in series, which the machine will reliably initiate. Current items are rated as 10-, 30-, 50-, and 100-cap capacity with the caps connected in a 500-foot-long firing line. The machine is used by attaching the firing line lead wires to the two terminal posts provided and then briskly actuating the machine handle.

## 2. Electric Firing Device

Electric firing devices serve the same purpose as blasting machines. They are, in general, of more recent origin and more varied in design and utility as indicated by the following general characteristics;

a. Most of the devices are powered by an internal battery and incorporate a capacitor. One developmental device, however, uses a blasting machine as the power source, and another uses an external battery pack as a supplementary power source to increase the output.

b. The devices usually incorporate a number of pairs of terminals and switches to permit the selective firing of a many as 10 firing circuits. One item, however, has no switches and one pair of terminals; firing is accomplished by attaching one of the firing line lead wires to a terminal then touching the other wire to the other terminal. The individual circuits of multicircuit devices can be internally connected in series or in parallel.

c. A self-test feature is usually incorporated in the form of an indicator lamp or voltmeter.

d. A circuit test feature is usually incorporated in the form of an indicator lamp or galvanometer.

## 3. Circuit Tester

The circuit tester is used to check the continuity of electric firing circuits. Two types, the blasting galvanometer and the electric test set, are used. The galvanometer measures circuit resistance while the electric test set indicates a continuous circuit by means of an indicator lamp. The electrical output of both types is necessarily restricted to less than the minimum required to initiate an electric blasting cap. The power source is an internal battery or a manually powered generator.

## 4. Firing Device

A firing device is essentially a fuze. Test procedures applicable to firing devices for mines and demolitions, and to similar items such as fuze igniters, are covered in TOP 4-2-505 which also descirbes the various types of firing devices.



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evaluation is normally the first phase of testing conducted; a preliminary safety evaluation, as a minimum, is required within 30 days after test initiation.

a. The safety evaluation of demolition-initiating equipment must demonstrate that the equipment is safe to transport and handle over a range of temperatures and that it is not subject to accidental initiation. The subtests that comprise the safety evaluation are paragraph 4.2.1 below, the 40-foot drop (TOP 4-2-601[5]), rough handling (TOP 4-2-602[6]), transportation vibration (TOP 1-2-601[7]), and radio frequency hazards (TOP 3-2-616[8]).

b. The developer is required by AR 385-16 to submit a safety assessment report before testing begins. A review of this report is mandatory. The purpose of the review is to identify design or performance characteristics that merit particular attention.

c. Review developmental test data to identify aspects of design or performance that deserve particular attention. Valid test results can be used to satisfy safety evaluation requirements or to supplement the results of limited additional tests.

d. Review the maintenance literature (usually a draft equipment publication). Particular emphasis is placed on the adequacy of any warning statements and adherence to established safe operating procedures.

4.2.1 Performance Safety Check. This subtest, the first phase of the safety evaluation, consists of taking a sample of the size shown in Table 1 and operating (or functioning) each of the items at ambient temperature. If no safety problem or hazard is encountered, the remainder of the safety evaluation is conducted.

TABLE 1 - PERFORMANCE SAFETY CHECK REQUIREMENTS

| Equipment Category | Minimum Sample | Para No. |
|---|---|---|
| Blasting machine | 6 | 4.3.1.1 |
| Electric firing device | 6 | 4.3.2 |
| Circuit tester | 6 | 4.3.3.2 |
| Electric blasting cap | 20 | 4.3.4.1 |
| | 20 | 4.3.4.2 |
| Nonelectric blasting cap | 20 | 4.3.5 |
| Time blasting fuse | 10 | 4.3.6.1 |
| | 10 | 4.3.6.2 |
| Detonating cord | 10 | 4.3.7.1 |
| | 10 | 4.3.7.2 |

4.3 Performance Tests.

4.3.1 Blasting Machines. Three types of firing performance tests are described below. Two use electric blasting caps incorporated in a firing circuit. The other, the simulated firing performance test, involves the use of an oscilloscope to display the electrical output of the blasting machine. The results of all three tests are considered in the durability test.



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firing. If the voltage trace is not equivalent, repeat the test in 4.3.1.1 before conducting additional simulated firings.

4.3.1.4 Durability Test. Subject each blasting machine undergoing this test to repeated firing attempts until it becomes inoperable or the number of successful operations (firing missions) required by the guidance document is achieved. Firings conducted by all of the previously described methods are considered. Most of the firing tests will necessarily be conducted by the simulated method. Each firing attempt made in the optional firing performance test 4.3.1.2 is considered to be a successful firing. Record the total number of successful firings and the cause of any failure, if determined.

4.3.2 Electric Firing Devices.

Circuit Tester Output. Measure the electrical output of each firing device in the circuit test mode by the method described in paragraph 4.3.3.

4.3.2.1 Circuitry Performance Test.

a. Prepare an electric firing circuit consisting of an electric blasting cap and 100 feet of two-conductor No. 18 AWG wire for each pair of firing device terminals.

b. Place all electric firing device switches in the OFF or SAFE position, and connect the firing circuits.

c. Check the continuity of each firing circuit with the integral circuit test mechanism. Correct any unsatisfactory circuit condition. Record the results of the circuit continuity check.

d. Individually fire and inspect each circuit after each firing to determine whether only the planned firing occurred. Record the inspection results.

e. Repeat steps a. through d. with each electric firing device.

4.3.2.2 Firing Performance Test. Perform at least two tests as described below with each electric firing device.

a. Prepare the maximum number of circuits that the device can fire simultaneously. The parameters of the firing circuits, with respect to number of electric blasting caps and length and type of wire, are designed to represent the maximum essential requirements expressed in the guidance document. Record the circuit parameters.

b. Connect the firing circuits and perform continuity checks as in 4.3.2.1.b and c above.

c. Fire all firing circuits simultaneously. Inspect for initiation of the blasting caps in each circuit, and record results.

4.3.2.3 Optional Firing Performance Test. The purpose of this optional test is to determine whether the electric firing device will initiate firing circuits that satisfy the desired requirements of the guidance document. When conducted, perform the test as in 4.3.2.2 above, except that:



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a. Tape a blasting cap to a length (at least 3 feet) of detonating cord. Place the blasting cap 6 inches from one end of the detonating cord with the explosive end of the cap toward the other end of the cord.

b. Initiate the blasting cap.

c. Inspect for complete detonation of the cord excluding the 6-inch-long section, and record the inspection results.

4.3.7.2 Propagation Test. Perform each of the following tests.

a. Square knot connection:

(1) Tie two 5-foot lengths of detonating cord together with a square knot. Place the connected lengths in a straight line. Tape a blasting cap to one end as described in 4.3.7.1.a.

(2) Initiate the blasting cap, and record whether or not propagation occurs across the knot.

b. Girth hitch connection:

(1) Place two lengths of detonating cord approximately 5-feet long perpendicular to each other; one is assumed to be a main line, the other a branch line. Connect the branch line to the main line with a girth hitch with one extra turn.

(2) Initiate the main line with a blasting cap. Record whether or not the detonation propagates from the main line to the branch line.

4.4 Extreme Operating Temperatures. If the test items perform satisfactorily (as well as being safe) at the extreme temperatures during the transportation-vibration phase of the safety evaluation, no further extreme-temperature test is needed. If performance shortcomings are observed, however, additional extreme-temperature tests are required in accordance with the guidance document. If temperatures are not shown, -37° C (-35° F) and 63° C (+145° F) are used if the intermediate conditions of AR 70-38[9] are specified. If, instead, the cold climate of AR 70-38 is specified, the low temperature test is conducted at -46° C (-50° F). If the hot-dry climate is specified, the high temperature test is conducted at 63° C on the assumption that the kit permitted by AR 70-38 consists of shade; if exposure to the open sun is required, a test temperature of 74° C (+165° F) will be necessary. In all cases, performance will be at the conditioning temperature and in accordance with the sample sizes and procedures shown in Table 1.

NOTE: The 63° C for intermediate conditions and the 74° C for hot-dry conditions take into account the effects of both air temperature and solar radiation but are only approximations. If a solar radiation facility is available, it is preferable to expose the test items to the diurnal cycles described in TOP 4-2-826[10] to obtain more precise temperatures.

4.5 Reliability and Durability. The reliability subtest and the durability subtest are usually combined and considered as one since the same data are used for both and since durability is actually a special case of reliability. The reliability and durability subtest not only uses the data from all of the other



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b. Connect the circuit tester to a variable resistor with a maximum resistance exceeding that which the guidance document requires the item to check. Step (1) follows if the circuit tester is of the galvanometer type. Step (2) follows if the circuit tester is of the indicator lamp type.

(1) Set the variable resistor at 10 ohms. Actuate the device and record the galvanometer reading. Repeat this step at each 10-ohm interval until a full-scale reading is obtained on the circuit tester. Record the galvanometer reading for each 10-ohm increment.

(2) Set the variable resistor at the maximum resistance value at which the guidance document requires the circuit tester to indicate a continuous circuit. Actuate the circuit tester. If the indicator lamp does not light on any of three consecutive actuations, reduce the resistance value by 10 ohms. If the indicator lamp lights on all three actuations, increase the resistance value by 10 ohms. Repeat this step until the maximum resistance value at which the lamp lights on three consecutive actuations is identified. Record this value.

4.3.3.3 Durability Test. Consider each previous actuation of devices subjected to this test. Perform additional actuations until the circuit tester fails or the number of actuations required by the guidance document is achieved. Perform the additional actuations with the circuit tester connected to a circuit having a resistance value of one-half that which previously produced a full-scale meter reading or three consecutive indicator lamp outputs. Record the number of successful circuit tests performed.

4.3.4 Electric Blasting Caps.

4.3.4.1 Nonfunctioning Test. The purpose of this test is to determine whether the electric blasting cap is subject to initiation by the maximum current level permissible for circuit testers. In this test, expose individual blasting caps to a current of 0.20 ampere for 5 seconds. Record any blasting cap initiations, and destroy nonfunctioning caps.

4.3.4.2 Individual Functioning Test. Test individual electric blasting caps as follows:

a. Prepare a test circuit capable of supplying a 0.45-ampere current to individual blasting caps.

b. Expose each blasting cap to the 0.45-ampere current for at least 10 seconds.

c. Measure and record the delay between current application and blasting cap firing to the nearest millisecond.

d. Destroy any blasting caps that fail to function.

4.3.4.3 Multiple Firing Test.

a. Prepare a firing circuit consisting of 500 feet of two-conductor No. 18 AWG firing wire and 10 series-connected electric blasting caps.



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b. Connect the circuit tester to a variable resistor with a maximum resistance exceeding that which the guidance document requires the item to check. Step (1) follows if the circuit tester is of the galvanometer type. Step (2) follows if the circuit tester is of the indicator lamp type.

(1) Set the variable resistor at 10 ohms. Actuate the device and record the galvanometer reading. Repeat this step at each 10-ohm interval until a full-scale reading is obtained on the circuit tester. Record the galvanometer reading for each 10-ohm increment.

(2) Set the variable resistor at the maximum resistance value at which the guidance document requires the circuit tester to indicate a continuous circuit. Actuate the circuit tester. If the indicator lamp does not light on any of three consecutive actuations, reduce the resistance value by 10 ohms. If the indicator lamp lights on all three actuations, increase the resistance value by 10 ohms. Repeat this step until the maximum resistance value at which the lamp lights on three consecutive actuations is identified. Record this value.

4.3.3.3 Durability Test. Consider each previous actuation of devices subjected to this test. Perform additional actuations until the circuit tester fails or the number of actuations required by the guidance document is achieved. Perform the additional actuations with the circuit tester connected to a circuit having a resistance value of one-half that which previously produced a full-scale meter reading or three consecutive indicator lamp outputs. Record the number of successful circuit tests performed.

4.3.4 Electric Blasting Caps.

4.3.4.1 Nonfunctioning Test. The purpose of this test is to determine whether the electric blasting cap is subject to initiation by the maximum current level permissible for circuit testers. In this test, expose individual blasting caps to a current of 0.20 ampere for 5 seconds. Record any blasting cap initiations, and destroy nonfunctioning caps.

4.3.4.2 Individual Functioning Test. Test individual electric blasting caps as follows:

a. Prepare a test circuit capable of supplying a 0.45-ampere current to individual blasting caps.

b. Expose each blasting cap to the 0.45-ampere current for at least 10 seconds.

c. Measure and record the delay between current application and blasting cap firing to the nearest millisecond.

d. Destroy any blasting caps that fail to function.

4.3.4.3 Multiple Firing Test.

a. Prepare a firing circuit consisting of 500 feet of two-conductor No. 18 AWG firing wire and 10 series-connected electric blasting caps.



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a. Tape a blasting cap to a length (at least 3 feet) of detonating cord. Place the blasting cap 6 inches from one end of the detonating cord with the explosive end of the cap toward the other end of the cord.

b. Initiate the blasting cap.

c. Inspect for complete detonation of the cord excluding the 6-inch-long section, and record the inspection results.

4.3.7.2 Propagation Test. Perform each of the following tests.

a. Square knot connection:

(1) Tie two 5-foot lengths of detonating cord together with a square knot. Place the connected lengths in a straight line. Tape a blasting cap to one end as described in 4.3.7.1.a.

(2) Initiate the blasting cap, and record whether or not propagation occurs across the knot.

b. Girth hitch connection:

(1) Place two lengths of detonating cord approximately 5-feet long perpendicular to each other; one is assumed to be a main line, the other a branch line. Connect the branch line to the main line with a girth hitch with one extra turn.

(2) Initiate the main line with a blasting cap. Record whether or not the detonation propagates from the main line to the branch line.

4.4 Extreme Operating Temperatures. If the test items perform satisfactorily (as well as being safe) at the extreme temperatures during the transportation-vibration phase of the safety evaluation, no further extreme-temperature test is needed. If performance shortcomings are observed, however, additional extreme-temperature tests are required in accordance with the guidance document. If temperatures are not shown, -37° C (-35° F) and 63° C (+145° F) are used if the intermediate conditions of AR 70-38[9] are specified. If, instead, the cold climate of AR 70-38 is specified, the low temperature test is conducted at -46° C (-50° F). If the hot-dry climate is specified, the high temperature test is conducted at 63° C on the assumption that the kit permitted by AR 70-38 consists of shade; if exposure to the open sun is required, a test temperature of 74° C (+165° F) will be necessary. In all cases, performance will be at the conditioning temperature and in accordance with the sample sizes and procedures shown in Table 1.

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firing. If the voltage trace is not equivalent, repeat the test in 4.3.1.1 before conducting additional simulated firings.

4.3.1.4 Durability Test. Subject each blasting machine undergoing this test to repeated firing attempts until it becomes inoperable or the number of successful operations (firing missions) required by the guidance document is achieved. Firings conducted by all of the previously described methods are considered. Most of the firing tests will necessarily be conducted by the simulated method. Each firing attempt made in the optional firing performance test 4.3.1.2 is considered to be a successful firing. Record the total number of successful firings and the cause of any failure, if determined.

4.3.2 Electric Firing Devices.

Circuit Tester Output. Measure the electrical output of each firing device in the circuit test mode by the method described in paragraph 4.3.3.

4.3.2.1 Circuitry Performance Test.

a. Prepare an electric firing circuit consisting of an electric blasting cap and 100 feet of two-conductor No. 18 AWG wire for each pair of firing device terminals.

b. Place all electric firing device switches in the OFF or SAFE position, and connect the firing circuits.

c. Check the continuity of each firing circuit with the integral circuit test mechanism. Correct any unsatisfactory circuit condition. Record the results of the circuit continuity check.

d. Individually fire and inspect each circuit after each firing to determine whether only the planned firing occurred. Record the inspection results.

e. Repeat steps a. through d. with each electric firing device.

4.3.2.2 Firing Performance Test. Perform at least two tests as described below with each electric firing device.

a. Prepare the maximum number of circuits that the device can fire simultaneously. The parameters of the firing circuits, with respect to number of electric blasting caps and length and type of wire, are designed to represent the maximum essential requirements expressed in the guidance document. Record the circuit parameters.

b. Connect the firing circuits and perform continuity checks as in 4.3.2.1.b and c above.

c. Fire all firing circuits simultaneously. Inspect for initiation of the blasting caps in each circuit, and record results.

4.3.2.3 Optional Firing Performance Test. The purpose of this optional test is to determine whether the electric firing device will initiate firing circuits that satisfy the desired requirements of the guidance document. When conducted, perform the test as in 4.3.2.2 above, except that:



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a. The safety evaluation of demolition-initiating equipment must demonstrate that the equipment is safe to transport and handle over a range of temperatures and that it is not subject to accidental initiation. The subtests that comprise the safety evaluation are paragraph 4.2.1 below, the 40-foot drop (TOP 4-2-601[5]), rough handling (TOP 4-2-602[6]), transportation vibration (TOP 1-2-601[7]), and radio frequency hazards (TOP 3-2-616[8]).

b. The developer is required by AR 385-16 to submit a safety assessment report before testing begins. A review of this report is mandatory. The purpose of the review is to identify design or performance characteristics that merit particular attention.

c. Review developmental test data to identify aspects of design or performance that deserve particular attention. Valid test results can be used to satisfy safety evaluation requirements or to supplement the results of limited additional tests.

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4.2.1 Performance Safety Check. This subtest, the first phase of the safety evaluation, consists of taking a sample of the size shown in Table 1 and operating (or functioning) each of the items at ambient temperature. If no safety problem or hazard is encountered, the remainder of the safety evaluation is conducted.

TABLE 1 - PERFORMANCE SAFETY CHECK REQUIREMENTS

| Equipment Category | Minimum Sample | Para No. |
|---|---|---|
| Blasting machine | 6 | 4.3.1.1 |
| Electric firing device | 6 | 4.3.2 |
| Circuit tester | 6 | 4.3.3.2 |
| Electric blasting cap | 20 | 4.3.4.1 |
| | 20 | 4.3.4.2 |
| Nonelectric blasting cap | 20 | 4.3.5 |
| Time blasting fuse | 10 | 4.3.6.1 |
| | 10 | 4.3.6.2 |
| Detonating cord | 10 | 4.3.7.1 |
| | 10 | 4.3.7.2 |

4.3 Performance Tests.

4.3.1 Blasting Machines. Three types of firing performance tests are described below. Two use electric blasting caps incorporated in a firing circuit. The other, the simulated firing performance test, involves the use of an oscilloscope to display the electrical output of the blasting machine. The results of all three tests are considered in the durability test.



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## 1. Blasting Machine

The blasting machine is a manually powered generator for initiating electric blasting caps. Blasting machines are rated according to the number of electric blasting caps, connected in series, which the machine will reliably initiate. Current items are rated as 10-, 30-, 50-, and 100-cap capacity with the caps connected in a 500-foot-long firing line. The machine is used by attaching the firing line lead wires to the two terminal posts provided and then briskly actuating the machine handle.

## 2. Electric Firing Device

Electric firing devices serve the same purpose as blasting machines. They are, in general, of more recent origin and more varied in design and utility as indicated by the following general characteristics;

a. Most of the devices are powered by an internal battery and incorporate a capacitor. One developmental device, however, uses a blasting machine as the power source, and another uses an external battery pack as a supplementary power source to increase the output.

b. The devices usually incorporate a number of pairs of terminals and switches to permit the selective firing of a many as 10 firing circuits. One item, however, has no switches and one pair of terminals; firing is accomplished by attaching one of the firing line lead wires to a terminal then touching the other wire to the other terminal. The individual circuits of multicircuit devices can be internally connected in series or in parallel.

c. A self-test feature is usually incorporated in the form of an indicator lamp or voltmeter.

d. A circuit test feature is usually incorporated in the form of an indicator lamp or galvanometer.

## 3. Circuit Tester

The circuit tester is used to check the continuity of electric firing circuits. Two types, the blasting galvanometer and the electric test set, are used. The galvanometer measures circuit resistance while the electric test set indicates a continuous circuit by means of an indicator lamp. The electrical output of both types is necessarily restricted to less than the minimum required to initiate an electric blasting cap. The power source is an internal battery or a manually powered generator.

## 4. Firing Device

A firing device is essentially a fuze. Test procedures applicable to firing devices for mines and demolitions, and to similar items such as fuze igniters, are covered in TOP 4-2-505 which also descirbes the various types of firing devices.



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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

AMSTE-RP-702-103
*Test Operations Procedure 4-2-045 26 February 1985
AD No. A151272

# DEMOLITION-INITIATING EQUIPMENT


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1. SCOPE. This TOP provides guidance for planning tests of demolition-initiating equipment to assure conformance with requirements documents. Tests to satisfy the requirements for the particular test item and test type can be selected from those listed in paragraph 4. This procedure does not apply to firing devices (fuzes) as defined in Appendix A, to initiating components of special demolition items or kits, or to their tools and accessories. This TOP does not cover mines and demolitions; for testing procedures regarding these items, see TOP 4-2-505.[1] **

Demolition-initiating equipment consists of the explosive and nonexplosive items used to initiate standard or expedient demolition explosive charges.

*This TOP supersedes TOP 4-2-045 dated 2 February 1972.

**Footnote numbers correspond to reference numbers in Appendix B.

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Blasting Devices
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Describes procedures for testing demolition-initiating equipment. Includes explosive and nonexplosive items used to detonate explosive charges.

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